

Cohort Profile

Cohort Profile: Brazilian High-Risk Cohort for Mental Health Conditions (BHRC)

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Key Features

- The Brazilian High-Risk Cohort for Mental Health Conditions was established in 2009 to investigate the developmental trajectories of mental health conditions, integrating clinical, social, genetic, neuroimaging, and environmental assessments. It is unique as one of the most comprehensive population neuroscience studies from a middle-income country.
- Participant screening was performed in 2009–10 across 57 public schools in Porto Alegre and São Paulo, Brazil, with 8012 caregivers reporting information on 9937 children aged 6–12 years.
- A total of 2511 participants were selected for genotyping and deep phenotyping: 958 randomly and 1553 selected by having a high risk of mental health conditions based on symptoms and family history.
- Participants received a baseline assessment in 2010–11 and were followed up at regular intervals, including four key waves, in 2013–14, 2018–19, 2019–20 (COVID-19), and 2023–25, with an 83% retention rate after 15 years.
- Currently, the cohort expanded to include three generations: parents, probands and probands' offspring, enabling transgenerational analyses of mental health and cognitive development across three generations.
- Data and resources are available through open-science platforms, with collaboration opportunities accessible via proposal submission (<https://osf.io/ktz5h/overview>).

Why was the cohort set up?

The Brazilian High-Risk Cohort for Mental Health Conditions (BHRC), formerly the High-Risk Cohort Study for Psychiatric Disorders in Childhood (HRC) [1], was established to advance the understanding of the developmental trajectories of pediatric mental health conditions by integrating information about behaviors, genes, environments, and brain development. The BHRC has an accelerated school-based longitudinal design in which children born between 1996 and 2004 were recruited at school in 2009 and 2010 and assessed every 2–4 years. The BHRC is one of the few population neuroscience studies from a middle-income country, integrating in-depth clinical assessments with data on environmental influences, genetics, imaging, cognition, and ecological momentary assessment (EMA), among others [2].

The cohort was established as a collaborative effort among three Brazilian universities: Universidade de São Paulo, Universidade Federal do Rio Grande do Sul, and Universidade Federal de São Paulo. This was possible through a networking grant from the Brazilian Science and Technology Ministry, which created the National Institute of Developmental Psychiatry for Children and Adolescents (in Portuguese, “Instituto Nacional de Psiquiatria do Desenvolvimento para a Infância e Adolescência”). Participants were recruited at state-funded schools of two Brazilian metropolitan areas: Porto Alegre (South, 1.5 million inhabitants, metropolitan area 4.4 million inhabitants) and São Paulo (Southeast, 11.4 million inhabitants, metropolitan area 22.8 million inhabitants).

Over the past 15 years, the study has been funded by multiple sources, including Conselho Nacional de Desenvolvimento Científico e Tecnológico, Conselho Nacional de Desenvolvimento Científico e Tecnológico, Fundação de Amparo à Pesquisa do Estado de São Paulo, European Research Council, UK Medical Research Council, Banco Industrial do Brasil S/A., and the Brazilian Ministry of Health. Additionally, the harmonization between the BHRC and other cohorts has been funded by the National Institutes of Mental Health (NIMH). The NIMH also provided support by mentorship from researchers from the Intramural Research Program to students working with BHRC data.

Who is in the cohort?

The BHRC enrolled participants from 57 state-funded schools (22 in Porto Alegre, 35 in São Paulo) during

compulsory registration days, on which at least one caregiver had to attend by law. Of the ~12 500 eligible caregivers, 8012 enrolled, providing information on 9937 children (participation: Porto Alegre, 65.1% and São Paulo, 63%). Inclusion criteria were registration in a state-funded school by a biological parent able to provide behavioral information and children aged 6–12 years. Parents were interviewed in person or by telephone, with a modified Family History Screen (FHS) administered by trained interviewers [1, 3]. The FHS was used to generate a family liability index that expressed the percentage of family members who screened positive for a mental health condition, adjusted for the degree of relatedness (mother, father, and sibling count as 1.0, half-sibling as 0.5). Most interviews involved biological mothers (87.3%), with the remainder completed by biological fathers. Based on the information reported at screening ($n = 9937$), we recruited two subgroups: random ($n = 958$) and high-risk ($n = 1553$) samples. Only one child per family was included. The high-risk sample was selected based on the presence of early mental health symptoms and high levels of symptoms in the family as assessed by using the family liability index [1]. Recruitment occurred from 28 September 2009 to 15 April 2010.

How often have they been followed up?

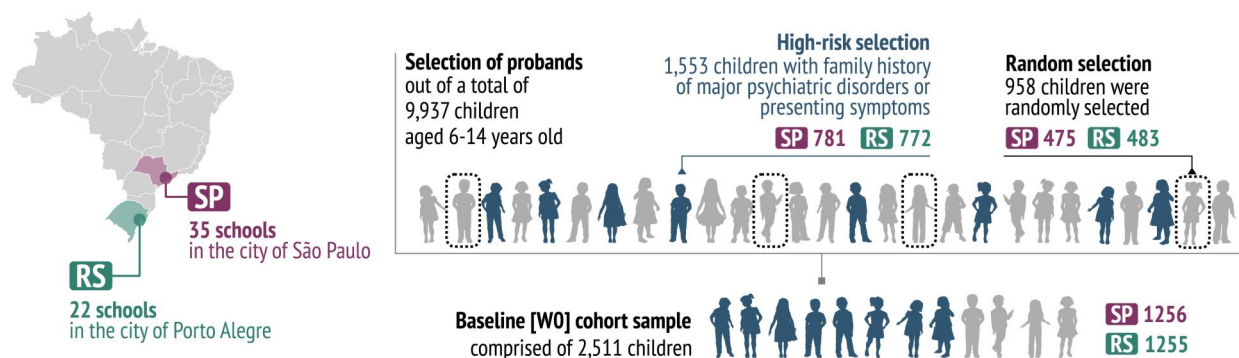
Participants have been reassessed over 15 years. Screening was undertaken in 2009–10, followed by baseline (Wave 0: August 2010–June 2011), Wave 1 (October 2013–November 2014), Wave 2 (November 2018–December 2019), and Wave 3 (2023–2025). A special follow-up (Wave C, April 2020–March 2021) evaluated the impact of COVID-19 (Fig. 1).

What has been measured?

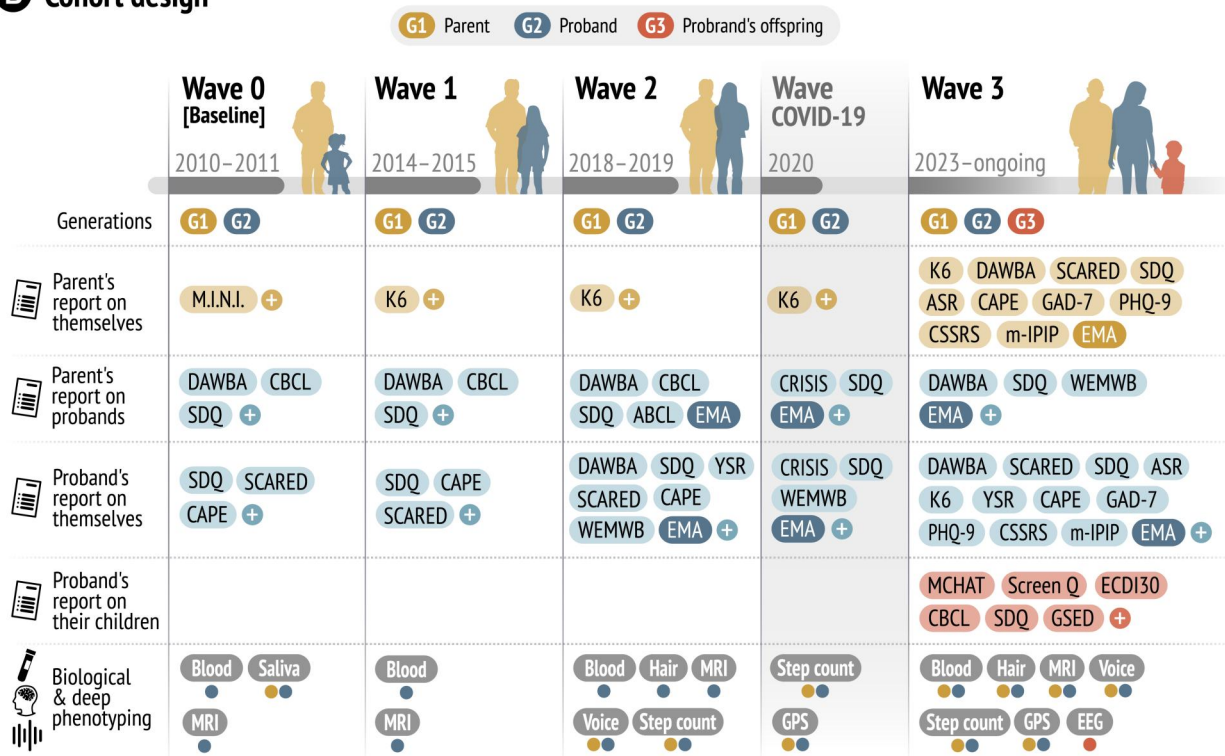
The BHRC collected a broad, harmonized set of measures to capture psychopathology, development, and context (Tables 1 and 2):

- Mental health data assessed via diagnostic interviews, rating scales, and parent, self, and teacher reports about psychopathology, temperament, and personality (Supplementary Table S1).
- Exposure to tobacco, alcohol, drugs, screen time, pornography, and addictive behaviors, along with their impact

A Probands' selection



B Cohort design



C Cohort profile

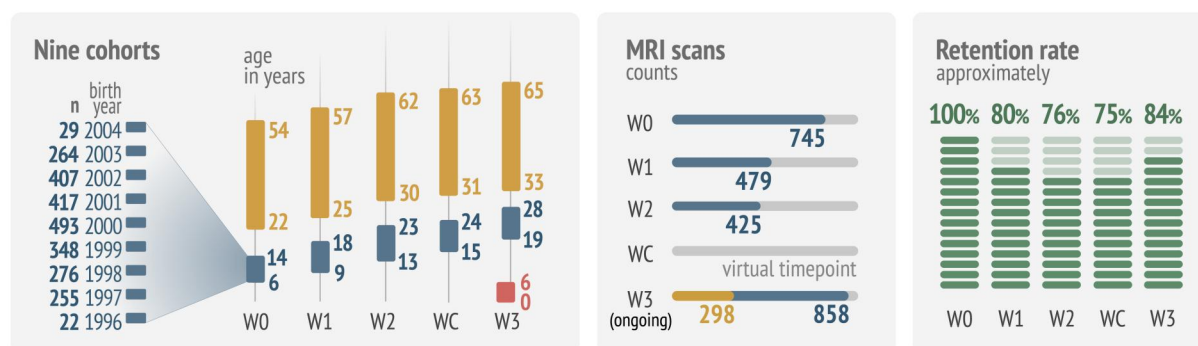


Figure 1. Study design. (A) Locations and sampling. RS, Rio Grande do Sul (Porto Alegre); SP, São Paulo. (B) Instruments and scales. ABCL, Adult Behavior Checklist; ASR, Adult Self-Report; CAPE, Community Assessment of Psychic Experiences; CBCL, Child Behavior Checklist; CRISIS, The Coronavirus Health Impact Survey; CSSRS, Columbia Suicide Severity Rating Scale; DAWBA, Development and Well-Being Assessment; ECDI-30, Early Childhood Development Index (30 items); GAD-7, Generalized Anxiety Disorder 7-item Scale; GSED, Global Scale for Early Development; K6, Kessler Distress 6-Item Scale; M-CHAT, Modified Checklist for Autism in Toddlers; MINI, Mini International Neuropsychiatric Interview; m-IPIP, Mini International Personality Item Pool; PHQ-9, Patient Health Questionnaire-9; SCARED, Screen for Child Anxiety Related Emotional Disorders; SDQ, Strengths and Difficulties Questionnaire; WEMWB/WEMWB, Warwick-Edinburgh Mental Well-Being Scale; YSR, Youth Self-Report. (C) Biological and digital measures. EEG, electroencephalography; GPS, global positioning system; MRI, magnetic resonance imaging.

Table 1. Behaviors, exposures, and service use.

Sections	Instruments
Mental health conditions assessed via diagnostic interviews, rating scales, and self-reports related to psychopathology, temperament, and personality (Supplementary Table S1)	Family history: FHS Psychiatric diagnosis: Developing and Well-Being Behavior Assessment (DAWBA), Mini International Neuropsychiatric Interview (M.I.N.I.), ICD-11 Prototype Matching Symptom questionnaires: Strengths and Difficulties Questionnaire, Kessler Distress 6-Item Scale (K6), Child Behavior Checklist (CBCL), Adult Behavior Checklist, Youth Self-Report, Adult Self-Report (ASR), Dimensional Yale Brown Obsessive Compulsive Scale, Hoarding Rating Scale-Interview (HRS-I), Community Assessment of Psychic Experiences, Comprehensive Assessment of At Risk Mental States, Screen for Child and Anxiety Related Emotional Disorders, Short Mood and Feeling Questionnaire, Patient Health Questionnaire (PHQ-9), Generalized Anxiety Disorder 7-item (GAD-7), Columbia Suicide Severity Rating Scale, Deliberate Self-Harm Inventory Temperament and personality: Early Adolescent Temperament Questionnaire—Revised, Mini International Personality Item Pool
Exposure to tobacco, alcohol, drugs, screen time, pornography, and addictive behaviors, and their impact on health and functioning (Supplementary Table S2)	Abuse and dependence: M.I.N.I. Screening: Alcohol, Smoking and Substance Involvement Screening Test Alcohol use: Alcohol (BHRC proper questionnaire), based on Alcohol Use Disorders Identification Test (AUDIT) Tobacco use: Tobacco (BHRC proper questionnaire), based on AUDIT Cannabis use: Driving after cannabis use (Global Drug Survey; GDS), Severity of Dependence Scale Drug use (BHRC proper questionnaire), based on Global Drug Survey Pornography: Problematic Pornography Consumption Scale
Cognitive assessments (Supplementary Table S3)	Basic information processing: Two-choice Reaction Time (2C-RT) Attention (alerting, orienting, conflict): Attention Network Attention orienting towards threats and rewards: Dot-probe (500ms and 1250ms) Temporal processing: Time anticipation (400 ms, 2 s), Duration Discrimination Delay aversion: Delay Reaction, Choice Delay Estimated intelligence quotient: Wechsler Intelligence Scale (vocabulary and block design) Premorbid intelligence quotient: Brazilian Word Accentuation Test Working memory: Digit Span (forward, backward), Corsi blocks (forward, backward) Inhibitory-based executive function: Conflict Control, Go/No-Go Nonverbal memory: Rey Osterrieth Complex Figure Test (copy, immediate, late) Visio-motor abilities: Bender Gestalt Test Fine motor abilities: Luria Motor Tasks Nonverbal design fluency: five-point task
Learning assessments (Supplementary Table S4)	School performance: School Performance Test, Reading Comprehension Test, Words/pseudowords Writing and Reading Test for Adults, CBCL—Academic Performance Section, ASR—Educational Section Auditory processes: Simplified Assessment of Auditory Processing, Phonological Awareness Instrument, The Phonology Test
Life events, functioning, lifestyle, and physical and sexual development (Supplementary Table S5)	Functioning: Warwick-Edinburgh Mental Well-Being, Job Satisfaction Scale, WHO Disability Assessment Schedule—Short, DAWBA—Friendship Section, ASR—Functioning Section Lifestyle, physical, and sexual development: Simple Physical Activity Questionnaire, Weight, Height, and abdominal circumference, Hours of physical inactivity, Pittsburgh Sleep Quality Index, Sexual orientation, gender identity, gender expression, sexual behavior (BHRC), Sexual experience (BHRC), Puberty status (BHRC), Duke University Religiosity Scale Life events: Traffic accident and consequences (PNS2009), Criminal contact, Diagnosis of chronic diseases (PNS2019), Diagnosis of sexually transmitted diseases (PNS2019)
Perinatal factors, stressful events and environments, bullying, stigma, and discrimination (Supplementary Table S6)	Demographics Perinatal and neonatal characteristics Stressful events and environments: Bullying at school (BHRC), Harassment at work (BHRC), Life events schedule (BHRC), Multi-informant Trauma Assessment, Multi-informant composite of childhood trauma, COVID-related changes (BHRC), The CoRoNaVirus Health Impact Survey, Juvenile Victimization Questionnaire (JVQ-R2), Peer Interaction subscale (Child Health and Illness Profile), Family Environment Scale, Social Cohesion/Neighbourhood Measure, Interviewer Observational Protocol Stigma and discrimination: Adapted Stigma Coping Measure, Reported and Intended Behaviour Scale (RIBS)
Service use and mental health literacy (Supplementary Table S7)	Service use: Services Assessment for Children and Adolescents, Client Service Receipt Inventory, Resource Generator—UK (Expert Advice subscale) Mental health literacy: Vignette-based Mental Health Literacy, Self-recognition questionnaire

Table 2. Genetics, imaging, and other protocols.

Genetic assessment from biospecimens (Supplementary Table S8)	Genomics Illumina Global Screening Array: G2: 2189 G1 mothers: 1898 G1 fathers: 1017 Illumina HumanCore Array: G2: 720 Whole-genome sequencing—30X: G2: 2293 G1 mothers: 712 Telomere length: G2: 560 at w0 G2: 366 at w1 G2: 1134 at w2 Mitochondrial DNA copy number: G2: 600 at w0 G2: 458 at w1 Transcriptomics Illumina HT-12 Beadchips: G2: 103 at w0 G2: 103 at w1 Inflammation genes reverse transcription polymerase chain reaction test (RT-PCR): G2: 130 at w0 Messenger RNA sequencing: G2: 600 at w0 G2: 474 at w1 Epigenomics Serum Extracellular Vesicle microRNA (EV-miRNA) sequencing: G2: 116 at w1 G2: 116 at w2 Serum EV-miRNA sequencing: G2: 500 EPIC array v1.0 (866 544 probes): G2: 24 at w0 G2: 24 at w1 EPIC array v2.0 (936 866 probes): G2: 369 at w0 G2: 371 at w1 G2: 326 at w2
Acquisition parameters for neuroimaging assessments (Supplementary Table S9)	Wave 0 GE Healthcare—Signa HDxt and Signa Excite—1.5T Sequence: T1, DTI, MT ON, MT OFF, Resting State N = 745 Wave 1 GE Healthcare—Signa HDxt and Signa Excite—1.5T Sequence: T1, DTI, MT ON, MT OFF, Resting State N = 479 Wave 2 GE Healthcare—Signa HDxt and Signa Excite—1.5T Sequence: T1, T2-FLAIR, DTI, Resting State N = 428 Wave 3 Philips—Achieva and Ingenia—3T & Siemens—Prisma Fit and Magnetom Vida—3T Sequence: T1, T2, field map P-A, field map A-P, Peer, Resting State 1, Movie 1, Resting State 2, Movie 2, Resting state 3 N = 1200 (<i>ongoing</i>)
EMA evaluation (Supplementary Table S10)	Intensive protocol—28 days after installing the app—once per day: Step count, Time spent with youth, Sleep (Motor Activity Research Consortium for Health [March] protocol) Intensive protocol—28 days after installing the app—five times a day: Overall distress (BHRC), Environment (Urban Mind), Mood and behavior (mMARCH; Mood Circumplex) Non-intensive protocol—up to 1 year after installing the app—weekly: Sleep (mMARCH), Strengths and Difficulties Questionnaire
Other protocols (Supplementary Tables S11–S17)	Supplementary Table S11—audio and video recordings: Wave 2 (N = 1610; 94.0%) and 3 (N = 1859; 95.2%) Supplementary Table S12—psychiatric interview: (A) free interview (20 min); (B) the Structured Clinical Interview for DSM-5 (SCID-5); (C) prototype matching for ICD11 Supplementary Table S13—interviews with participants who are not able to inform about their mental health concerns Supplementary Table S14—technical specifications of investigating drug substances in the hair Supplementary Table S15—technical specifications of investigating drug substances in the hair Supplementary Table S16—consortium papers and harmonization with other cohorts Supplementary Table S17—technical specifications of the protocol of the new offspring cohort (cohort members with children = 458 in November 2024, G3)

G1, Generation 1, parents of the main probands; G2, Generation 2, main probands; w0, Wave 0 or baseline; w1, Wave 1 or 3-year follow-up; w2, Wave 2 or 8-year follow-up; EV, extracellular vesicles.

on health and functioning via both self-reports and toxicological hair assays ([Supplementary Table S2](#)).

- Cognitive assessments, including intelligence quotient, executive function, memory, attention, and attention bias to threats and rewards ([Supplementary Table S3](#)).
- Learning assessments, including reading speed, accuracy and comprehension, writing, and mathematics ([Supplementary Table S4](#)).
- Life events, functioning, and lifestyle, including friendship, sleep patterns, well-being, religiosity, physical

health, traffic accidents, exposure to violence, and medical diagnoses ([Supplementary Table S5](#)).

- Perinatal factors, stressful events and environments, bullying, stigma, and discrimination ([Supplementary Table S6](#)).
- Service use, including inpatient admission, outpatient services, help-seeking behavior, treatments, and barriers encountered in accessing care ([Supplementary Table S7](#)).
- Genetic assessment from biospecimens (saliva and blood), including genomics, transcriptomics, and epigenomics ([Supplementary Table S8](#)).

- Neuroimaging assessment, including diverse modalities of structural and functional magnetic resonance imaging ([Supplementary Table S9](#)).
- EMA, including mood, activities, step counts, GPS location, and time spent in nature ([Supplementary Table S10](#)).

Quality control

From the 8-year follow-up onwards (Wave 2), all interviews were audio-recorded ([Supplementary Table S11](#)) and reviewed for quality control. At the 15-year follow-up, the protocol was further enhanced by including an additional psychiatric interview tailored to severe conditions, i.e. intellectual disability, imprisonment, or psychotic disorder ([Supplementary Tables S12 and S13](#)).

Assessing biological parents (G1)

Biological parents (Generation 1, G1) initially served as informants about their children (G2). As the cohort matured and the G2 participants reached adulthood, G1 participation was expanded: parents were interviewed about their own health, education, and functioning by using the same core protocol as was employed for their offspring. Including both biological parents where possible effectively tripled the number of assessed individuals and enabled family-based analyses (e.g. triads) that were crucial for disentangling genetic, environmental, and intergenerational mechanisms.

Adding a third generation (G3)

The 15-year follow-up introduced the third generation (G3): the children of the original G2 participants. By November 2024, 458 G2 participants had had at least one child, yielding 574 G3 children enrolled. Initial contact identified the child's main caregiver to complete an online sociodemographic and perinatal questionnaire, followed by a recorded online interview by a trained psychologist that focused on the child's emotions, behaviors, and environment ([Supplementary Table S17](#)). Adding G3 has allowed the cohort to investigate trans-generational associations and inherited susceptibilities with direct measures across three generations.

What has it found?

Mental health conditions are frequent, fluctuate over time, and most Brazilian children with such conditions do not receive needed care

One in five children in the studied cities experiences a mental disorder [1], yet only 20% of those receive any form of treatment [4, 5], with parental stigma acting as a major barrier to accessing care [6]. Trajectories show waxing and waning patterns of symptoms, with both homotypic and heterotypic transitions. Externalizing conditions, such as attention-deficit hyperactivity disorder (ADHD) and disruptive behavior disorder, were more prevalent in earlier waves, whereas anxiety and depression were more prevalent in later waves ([Table 3](#)).

Not only symptoms, but positive traits have an important role in educational outcomes

Problems such as low test performance, age-grade distortion, and retention were associated with psychopathology [7]. Positive traits such as eagerness to learn, affection, and caring play a significant role in mitigating negative educational outcomes [8]. They buffer the effects of lower intelligence on

learning and reduce the impact of psychopathology on academic performance.

Social determinants and childhood adversity influence mental health, brain activity, and development

Poverty, affecting 12% of the sample, is associated with externalizing conditions [9] and later-life criminal convictions [9]. Socioeconomic correlated with spontaneous activity in the right superior temporal gyrus, part of the extended language network [10]. Adversity showed distinct effects: threat-related experiences were tied to psychopathology, while deprivation impaired executive function [11]. Traumatic events with intentional harm were more strongly associated with psychotic experiences than non-intentional trauma [12]. Parental job loss worsens children's mental health, increasing clinical diagnoses [13].

ADHD involves deficient information processing, biased attention orienting to threats characterizes both anxiety and irritability, and executive functions relate to general psychopathology

While deficits in basic information processing (e.g. processing speed and stimulus discriminability) were specific to ADHD [14], they were also present in children with subclinical inattention symptoms, providing support for ADHD dimensionality [15]. Impaired basic information processing mediated the relationship between ADHD symptoms and poor reading ability, particularly in younger children [16]. Biases in attention orienting were associated with irritability in children and adolescents [17], beyond their association with anxiety. Finally, impairments in executive functioning were linked to general psychopathology [17], rather than specific disorders.

Psychopathology has many dimensions with distinct external correlates

We found psychopathology to have a hierarchical structure, with general and specific dimensions [19]. For example, anxiety disorders can be divided into fear and distress disorders, with distinct cognitive underpinnings [19] and familial transmission patterns [18]. Oppositionality comprises three main dimensions: irritability, headstrong behavior, and hurtfulness, each associated with unique familial and developmental outcomes [20]. Mania has two dimensions: exuberant and undercontrolled, with the latter linked to the most detrimental outcomes [21].

Data-driven classifications of executive function have strong predictive validity

We proposed a data-driven classification of executive dysfunction by using cognitive tasks assessing inhibitory control, working memory, and temporal processing [22]. Participants with executive dysfunction reported symptoms spanning several domains of psychopathology and exhibited impairment in multiple settings, including more adverse school events. They also exhibited smaller intracranial volumes and reduced cortical surface areas.

Alzheimer's genetic risk has implications for early development

Genetic risk for Alzheimer's disease may affect early-life cognition and hippocampal volumes, supporting previous evidence that some forms of late-life dementia could have

Table 3. Prevalence rates of mental conditions across waves in random and high-risk groups.

Characteristic ^a	Wave 0			Wave 1			Wave 2		
	R ^b	HR ^c	W ^d	R ^b	HR ^c	W ^d	R ^b	HR ^c	W ^d
	N = 958	N = 1533		N = 768	N = 1242		N = 693	N = 1108	
Any disorder	191 (20%)	461 (30%)	19%	142 (18%)	329 (26%)	18%	159 (23%)	334 (30%)	23%
Any anxiety	84 (8.8%)	189 (12%)	7.5%	76 (9.9%)	191 (15%)	9.9%	93 (13%)	178 (16%)	12%
GAD	17 (1.8%)	30 (1.9%)	1.2%	8 (1.0%)	32 (2.6%)	1.0%	31 (4.5%)	63 (5.7%)	4.1%
Panic disorder	0 (0%)	1 (<0.1%)	<0.1%	4 (0.5%)	16 (1.3%)	0.6%	33 (4.8%)	54 (4.9%)	3.8%
Agoraphobia	0 (0%)	4 (0.3%)	<0.1%	0 (0%)	7 (0.6%)	0.3%	14 (2.0%)	32 (2.9%)	2.1%
Social phobia	3 (0.3%)	23 (1.5%)	0.6%	17 (2.2%)	49 (3.9%)	2.3%	20 (2.9%)	36 (3.3%)	2.6%
Specific phobia	28 (2.9%)	61 (3.9%)	2.6%	20 (2.6%)	43 (3.5%)	2.4%	6 (0.9%)	20 (1.8%)	1.0%
PTSD	8 (0.8%)	15 (1.0%)	0.6%	6 (0.8%)	7 (0.6%)	0.5%	17 (2.5%)	29 (2.6%)	2.0%
OCD	2 (0.2%)	5 (0.3%)	0.2%	3 (0.4%)	21 (1.7%)	0.7%	5 (0.7%)	13 (1.2%)	1.0%
Separation anxiety	20 (2.1%)	52 (3.3%)	1.5%	34 (4.4%)	77 (6.2%)	4.1%	5 (0.7%)	9 (0.8%)	0.6%
Other anxiety	15 (1.6%)	29 (1.9%)	1.5%	0 (0%)	2 (0.2%)	0.2%	1 (0.1%)	1 (<0.1%)	<0.1%
Any depression	12 (1.3%)	69 (4.4%)	1.6%	33 (4.3%)	106 (8.5%)	4.2%	80 (12%)	188 (17%)	12%
Major depression	11 (1.1%)	62 (4.0%)	1.3%	27 (3.5%)	88 (7.1%)	3.7%	78 (11%)	186 (17%)	12%
Other depression	1 (0.1%)	7 (0.5%)	0.2%	6 (0.8%)	18 (1.4%)	0.5%	2 (0.3%)	2 (0.2%)	0.2%
Any ADHD	79 (8.2%)	195 (13%)	8.2%	29 (3.8%)	76 (6.1%)	3.2%	14 (2.0%)	38 (3.4%)	2.3%
ADHD inattentive	24 (2.5%)	71 (4.6%)	2.7%	8 (1.0%)	29 (2.3%)	0.9%	8 (1.2%)	15 (1.4%)	1.1%
ADHD hyp-imp	13 (1.4%)	27 (1.7%)	1.2%	4 (0.5%)	3 (0.2%)	0.4%	0 (0%)	2 (0.2%)	<0.1%
ADHD combined	29 (3.0%)	76 (4.9%)	2.9%	6 (0.8%)	28 (2.3%)	0.9%	4 (0.6%)	9 (0.8%)	0.5%
Other hyperactivity	13 (1.4%)	21 (1.4%)	1.3%	11 (1.4%)	16 (1.3%)	1.0%	2 (0.3%)	12 (1.1%)	0.7%
Any DBD	53 (5.5%)	118 (7.6%)	5.0%	33 (4.3%)	73 (5.9%)	4.2%	17 (2.5%)	48 (4.4%)	3.0%
ODD	39 (4.1%)	92 (5.9%)	4.0%	19 (2.5%)	32 (2.6%)	1.8%	2 (0.3%)	8 (0.7%)	0.3%
Conduct disorder	11 (1.1%)	29 (1.9%)	0.9%	15 (2.0%)	40 (3.2%)	2.3%	15 (2.2%)	40 (3.6%)	2.7%
Other disruptive	6 (0.6%)	3 (0.2%)	0.3%	1 (0.1%)	2 (0.2%)	0.2%	0 (0%)	0 (0%)	0%
Any other	10 (1.0%)	38 (2.4%)	0.9%	16 (2.1%)	22 (1.8%)	1.4%	9 (1.3%)	20 (1.8%)	1.2%
Mania/bip	3 (0.3%)	2 (0.1%)	0.2%	0 (0%)	1 (<0.1%)	<0.1%	3 (0.4%)	2 (0.2%)	0.3%
Psychosis	0 (0%)	1 (<0.1%)	<0.1%	1 (0.1%)	1 (<0.1%)	<0.1%	1 (0.1%)	2 (0.2%)	0.1%
Eating disorder	2 (0.2%)	8 (0.5%)	0.2%	7 (0.9%)	11 (0.9%)	0.7%	5 (0.7%)	13 (1.2%)	0.7%
Autism	2 (0.2%)	13 (0.8%)	0.3%	0 (0%)	0 (0%)	0%	0 (0%)	0 (0%)	0%
Tic disorder	3 (0.3%)	16 (1.0%)	0.2%	8 (1.0%)	9 (0.7%)	0.7%	1 (0.1%)	3 (0.3%)	0.1%
Current DSH (Parent w0:2/Child w1:2)	10 (1.0%)	18 (1.2%)	0.7%	19 (2.6%)	43 (3.7%)	3.0%	55 (9.0%)	122 (12%)	9.3%

^a ADHD, attention-deficit hyperactivity disorder; bip, bipolar; DBD, disruptive behavior disorder; DSH, deliberate self-harm; GAD, generalized anxiety disorder; PTSD, post-traumatic stress disorder; OCD, obsessive-compulsive disorder; ODD, oppositional-defiant disorder.

^b Random, *n* (%).

^c High-risk, *n* (%).

^d Weighted, % (weighted via propensity score weighting using as weighting variables age, sex, number of siblings, number of symptoms, and family history of psychopathology).

developmental roots [23]. We also showed that this genetic risk influences functional connectivity among brain regions susceptible to tau pathology during childhood and adolescence [24].

Aggregated genetic risks have small but consistent effects, which are dependent on the environment

Although the effects of individual genetic variants are modest, their aggregation holds substantial predictive power, especially when moderated by environmental risks [25]. Different polygenic scores appear to impact cognition and brain volume in early development [23], as well as functional connectivity [24] and inhibitory control [22], among other traits. In addition, we found that parents' genetics influence the cognition and educational performance of the offspring—a relationship likely mediated by the parents' phenotypes [26].

Neurodevelopmental correlates of psychopathology include delayed default mode network maturation, cerebellum volumes smaller than expected for age, and aberrant ventral striatum connectivity

General child psychopathology was associated with delayed maturation of the default mode network using resting state

functional connectivity [27], while smaller-than-expected cerebellar volumes for a given age and sex were associated with higher externalizing psychopathology and worse executive function [28]. Aberrant ventral striatum functional connectivity predicted future risk for depressive disorder [29]—a finding replicated in an independent sample [30]. Nevertheless, imaging studies revealed relatively low effect sizes, with an aggregate measure of structural brain measures explaining no more than 1%–3% of the variance in behavioral traits [31].

Aggregating data is critical for scientific exploration

The BHRC data have contributed to numerous important consortia, addressing key gaps in mental health and neurodevelopmental research. For example, they contributed to showing that the diagnosis of ADHD in younger children within a classroom is no more likely to be disconfirmed over time than that of older children [32]; to the discovery of genes related to post-traumatic stress disorder [33]; and to charting brain trajectories of typical development across the lifespan [34]. This is particularly important because most genetic and imaging studies originate from high-income countries, leaving populations from low- and middle-income countries underrepresented. By contributing data to large consortia such as the Psychiatric Genomics Consortium, Enhancing NeuroImaging

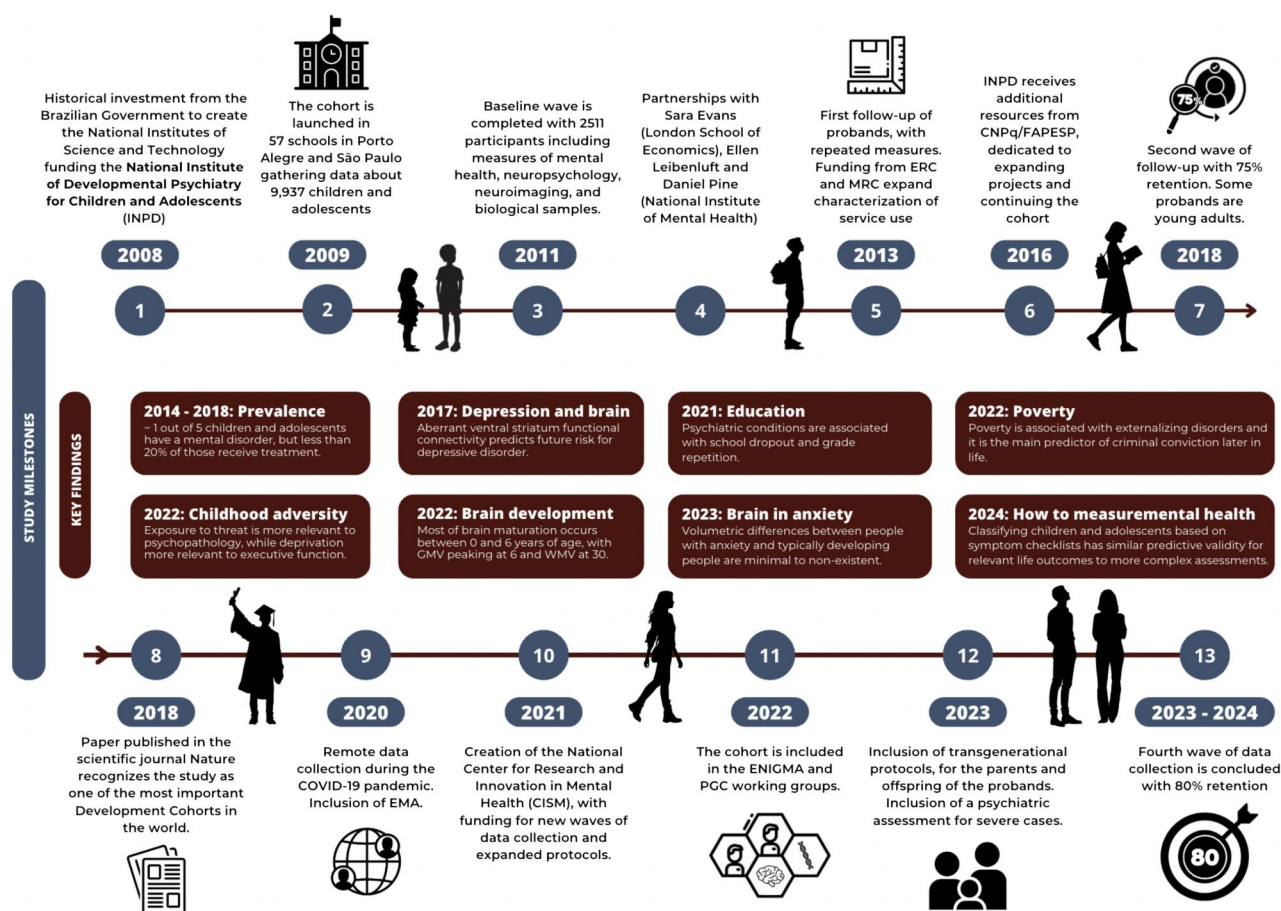


Figure 2. Milestones and major findings.

Genetics through Meta-Analysis, Reproducible Brain Charts [35], The Methylation, Imaging and NeuroDevelopment [36], and others (Supplementary Table S17), the BHRC has ensured that findings from Brazil add diversity and depth to global datasets (Fig. 2).

What are the main strengths and weaknesses? Strengths

The BHRC is notable for its comprehensive assessment of mental health phenotypes, encompassing dimensional and diagnostic approaches. Its long-term longitudinal design allows the study of both typical and atypical neurodevelopmental trajectories, leveraging a wide range of data modalities. These include biological samples (hair, blood, and saliva), genomic (single nucleotide polymorphism (SNP)-array, whole-genome sequencing, and telomere length), transcriptomic (RNA sequencing), and epigenetic data (DNA methylation array), along with environmental, neuroimaging, physical health, cognitive, and educational assessments. This cohort is particularly notable for its depth of phenotyping within the context of environmental and genomic diversity, reflecting Brazil's highly mixed genetic population [37]. As one of the few extensive cohorts from a low- and middle-income country, the BHRC has achieved a high retention rate of 83%. Analyses of baseline characteristics indicated that attrition across waves was generally limited: respondents and nonrespondents did not differ substantially in socioeconomic status or school grade, although maternal education consistently

predicted nonresponse, with children of less-educated mothers more likely to be lost to follow-up (Supplementary Table S18). Attrition was also higher among participants with baseline anxiety disorders, whereas other baseline psychiatric conditions did not show consistent differences. Additionally, at Wave 3, a higher proportion of nonrespondents were male compared with baseline participants. Importantly, potential selection biases at study entry were previously examined and reported [1], with some differences observed in the high-risk stratum but no systematic biases in the random-selection group. This suggests that selection bias is unlikely to have substantially influenced the results overall, although the differential attrition by sex, maternal education, and anxiety should be acknowledged.

The cohort had generated 147 publications as of August 2025. A complete list of publications is available at <https://osf.io/ktz5h/wiki/Bibliographic%20repository/>. Most importantly, the study contributed to capacity building and human resources, with many publications led by students and early-career researchers, including 13 by master's students, 65 by PhD students, 11 by undergraduate research assistants, and 12 by postdoctoral fellows.

Weaknesses

The sample is non-probabilistic and based on urban state-funded schools, which limits the generalizability of findings to the broader population. Additionally, while the primary focus of the BHRC on mental health has enabled in-depth study of psychiatric conditions, this emphasis limits

investigations into physical health. Finally, our functional neuroimaging protocol used in Waves 0, 1, and 2 (established in 2009) may not meet the current reliability standards required for functional analyses.

Can I get hold of the data? Where can I find out more?

The BHRC is committed to open science. De-identified data can be requested via a brief proposal through the project's Open Science Framework page (<https://osf.io/ktz5h/>). Upon approval by the principal investigators, data are shared securely. A detailed data dictionary, protocol documentation, wikis, and data-cleaning code are available to facilitate secondary analyses. Researchers interested in collaborating on the project are encouraged to contact Giovanni A. Salum (gsalumjr@gmail.com) and Pedro M. Pan (pedromariopan@gmail.com). Some of our imaging data are also readily available at <https://reprobrainchart.github.io>.

Ethics approval

The study was approved by the Brazilian National Ethics in Research Commission (CAAE 13852413010015327 and 74563817.7.1001.5327). For screening, Wave 0, Wave 1, and Wave 2, written consent was obtained from parents and, when appropriate, assent or consent from participating children. For those unable to read or write, verbal agreement was documented after a researcher had explained the procedures and answered questions. Families could schedule appointments with study psychologists and social workers to receive results and referrals when indicated. By Wave 3, participants had reached the legal age of consent; consent was obtained directly from them. For participants lacking legal capacity, consent was obtained from parents/guardians with verbal assent from the participant. Situations involving risk of harm received special attention that was consistent with ethical guidelines.

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Author contributions

Conceptualization: G.A.S., L.A.R., E.C.M., R.A.B., S.B., P.M.P. Data curation: A.S., A.Z., L.T.I., M.S. Formal analysis: A.S. Funding acquisition: G.A.S., M.P.M., T.S., S.E.-L., A.J., C.M., R.G.-O., A.C., A.T., M.S., L.A.R., J.M., E.C.M., R.A.B., S.B., P.M.P. Investigation: G.A.S., L.A.R., E.C.M., J.M., R.A.B., S.B., P.M.P. Methodology: G.A.S., L.A.R., E.C.M., R.A.B., S.B., P.M.P. Project administration: G.A.S., C.d.G., L.So., J.J., R.C., L.Su., A.C., A.S., P.B.R., S.B., R.A.B., L.A.R., E.C.M., P.M.P. Resources: G.A.S., M.P.M., T.S., S.E.-L., A.J., C.M., R.G.-O., A.C., A.T., M.S., L.A.R., E.C.M., R.A.B., S.B., P.M.P. Software: A.S., M.S., L.T.I. Supervision: G.A.

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Supplementary data

Supplementary data is available at *IJE* online.

Conflict of interest

A.C. declares receiving consulting fees from Knight Therapeutics and EMS pharmaceuticals in the last 3 years, unrelated to the current study. A.Z. is supported by the National Institute of Mental Health Intramural Research Program (ZIA-MH002781); this manuscript was prepared or accomplished by A.Z. in his personal capacity; the opinions expressed in this article are the author's own and do not reflect the view of the National Institutes of Health, the Department of Health and Human Services, or the US government. G.V.P. has served as a speaker and or consultant to Abbott, Aché, Adium, Apsen, EMS, Libbs, and Medice, and receives authorship royalties from Manole Editors. L.A.R. has received grant or research support from, served as a consultant to, and served on the speakers' bureau of Abdi Ibrahim, Abbott, Aché, Adium, Apsen, Bial, Celler, EMS, Hypera Pharma, Knight Therapeutics, Libbs, Medice, Novartis/Sandoz, Pfizer/Upjohn/Viatris, Shire/Takeda, and Torrent in the last 3 years. The ADHD and Juvenile Bipolar Disorder Outpatient Programs chaired by L.A.R. have received unrestricted educational and research support from the following pharmaceutical companies in the last 3 years: Novartis/Sandoz and Shire/Takeda. L.A.R. has received authorship royalties from Oxford Press and ArtMed. P.M.P. received payment or honoraria for lectures and presentations in educational events for Abbot, Libbs, Germed Pharma, Instituto Israelita de Pesquisa e Ensino Albert Einstein, and Instituto D'Or de Pesquisa e Ensino. All the other authors declare no conflict of interest.

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Data availability

De-identified data can be requested via a brief proposal through the project's Open Science Framework page (<https://osf.io/ktz5h/>).

Use of artificial intelligence (AI) tools

AI tools were not used in this manuscript.

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